Amrita Vishwa Vidyapeetham
School of Computing, Amritapuri.
Department of Computer Science and Engineering

Course Plan

1. Course Information

Course Code	**23CSE498**	**Title**	**PROJECT - PHASE – 2**
Academic Year	2026 – 2027 ODD	**Semester**	VII
Program	2023 BTech CSE	**L – T – P – C**	**0-0-12-6**

2. Faculty Information:

Sl. No	Faculty Name	Class
1	Dr. Aswathy Mohan	CSE A
2	Ms. Smriti Govind	CSE B
3	Dr. Tintu Vijayan	CSE C
4	Dr. Divya Udayan J [Course Mentor]	CSE D

3. Couse Objectives

- In this phase, the emphasis is on fine-tuning the problem statement through the insights obtained from the identification of any existing research gaps.
- This phase involves providing a detailed representation of the solution strategy, including modules and workflow, as well as designing the solution in terms of algorithms, diagrams, or models.
- This phase exposes the students to design-develop-debug cycle.
- The implementation of the solution is carried out in this phase.

4. Course Outcomes

CO1: Apply the insights from the problem analysis and literature survey to formalize the problem statement.

CO2: Design effective solution(s) for the identified problem.

CO3: Implement and verify the designed solution(s).

5. CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2		2				3	3	3		3	3	3
CO2	3	3	3	2	3			3	3	3	2	3	3	3
CO3	3	3	3		3			3	3	3		3	3	3

1 – Weak affinity, 2 – Moderate affinity, 3 – Strong affinity

6. CO-PO/PSO Affinity Justification

CO	PO/PSO	Affinity	Justification
CO1: Apply the insights from the problem analysis and literature survey to formalize the problem statement.	PO1	2	Students moderately apply engineering and computing fundamentals to understand the application domain and formulate a well-defined problem statement.
	PO2	2	Students analyze existing research, identify research gaps, and evaluate technical constraints to refine the problem definition.
	PO4	2	Literature survey and critical analysis of published research provide moderate exposure to investigation and evidence-based problem formulation.
	PO8	3	Strong emphasis is placed on research ethics, plagiarism-free literature survey, and proper citation of scholarly work.
	PO9	3	Students collaborate effectively with teammates and guides while reviewing literature and identifying research gaps.
	PO10	3	Preparation of the literature survey and project proposal enhances technical writing and communication skills.
	PO12	3	Students demonstrate lifelong learning by independently exploring recent research papers and emerging technologies.
	PSO1	3	Strong alignment through domain-specific problem identification based on current research trends.
	PSO2	3	Students extensively study recent technologies, methodologies, and tools

			relevant to their specialization before finalizing the problem.
CO2: Design effective solution(s) for the identified problem.	PO1	3	Students strongly apply engineering principles and computing knowledge to design an effective solution architecture.
	PO2	3	Analytical skills are used to compare alternative approaches and select appropriate algorithms and methodologies.
	PO3	3	Students design complete system architecture, workflows, algorithms, and models that satisfy the project requirements.
	PO4	2	Design decisions are supported through analysis of existing solutions and evaluation of different design alternatives.
	PO5	3	Students effectively utilize modern engineering tools, software frameworks, modeling tools, and development platforms during system design.
	PO8	3	Ethical considerations, originality of design, and responsible engineering practices are incorporated into the solution design.
	PO9	3	Strong teamwork is demonstrated while developing modular designs and integrating different solution components.
	PO10	3	Students communicate their solution through design documents, diagrams, algorithms, and technical discussions.
	PO11	2	Students moderately apply project planning, task allocation, and resource management while designing the solution.

	PO12	3	Students continuously acquire new knowledge by exploring advanced design techniques and emerging technologies.
	PSO1	3	Strong alignment through the development of domain-specific solution architectures.
	PSO2	3	Specialized software tools, frameworks, and technologies are employed to design the proposed solution.
CO3: Implement and verify the designed solution(s).	PO1	3	Students apply engineering knowledge and programming skills to implement the designed solution effectively.
	PO2	3	Students analyze implementation challenges, debug issues, and optimize system performance.
	PO3	3	Complete implementation and verification of the designed solution demonstrate strong engineering design capability.
	PO5	3	Students extensively use modern software development tools, programming environments, debugging utilities, and testing frameworks.
	PO8	3	Ethical software development practices, originality of implementation, and responsible use of resources are maintained.
	PO9	3	Team members collaborate during coding, integration, debugging, and validation activities.
	PO10	3	Students effectively communicate implementation progress, testing results, and technical findings through demonstrations and documentation.

	PO12	3	Students independently learn new technologies and development frameworks required for successful implementation.
	PSO1	3	Strong alignment as students implement domain-specific solutions addressing real-world problems.
	PSO2	3	Specialized tools, libraries, and platforms relevant to the discipline are effectively used during implementation and verification.

7. Project Evaluation Pattern

7.1 Evaluation Pattern

Table 7.1.1- Project Evaluation pattern

Internal	Proposal + Literature Survey	10
	Design and solution	10
	Progress review	20
	Mid Review	30
Total Internal		**70**
External	Final Review	20
	Project Report Submission	10
Total External		**30**

Sl. No	Component	Marks	Date	Mode
1	Proposal + Literature Survey(viva)	10	03-07-26	Submission
2	Design and solution(viva)	10	31-07-26	Submission
3	Progress review (Code)	20	07-09-26 to 11-09-26	Offline
4	Mid Review	30	22-09-26 to 28-09-26	Offline
5	Final Review	20	12-10-26 to 16-10-26	Offline
6	Project Report Submission	10	13-11-26	Submission

7.2 Process to assess individual and team performance

Table 7.2.1- Proposal Submission Rubrics and CO Mapping

Proposal Submission (10 Marks)		**Granular PO mapping**	**CO**
Proposal	5	PO1, PO2, PO4	CO2
Literature Survey	5	PO1, PO2, PO8, PO10	CO1

Table 7.2.2- Design Solution Rubrics and CO Mapping

Design and Solution(10)		**Granular PO mapping**	**CO**
Block diagram/architecture	5	PO2, PO3, PO5	CO2
Clarity in Writeup	5	PO3, PO10	CO1

Table 7.2.3- Progress Review Rubrics and CO Mapping

Code Review (20)		**PO**	**CO**
Design to Code Mapping	6	PO1, PO2, PO3, PO5, PO6, PO8, PO9, PO10, PO11, PO12	CO2
Code Explanation & Individual Contribution	14	PO1, PO2, PO4, PO5, PO8, PO9, PO10, PO11, PO12	CO3

Table 7.2.4- Mid Review Rubrics and CO Mapping

Mid Review (30)		**PO**	**CO**
Presentation	10	PO1, PO2, PO3, PO5, PO9, PO10, PO11, PO12	CO2

Viva	10	PO1, PO2, PO3, PO4, PO5, PO8, PO9, PO10, PO12	CO3
Result Analysis	10	PO1, PO2, PO4, PO5, PO9, PO10, PO11, PO12	CO3

Table 7.2.5- Final Review and Report Rubrics and CO Mapping

Component	Evaluation Criteria	Poor Performance	Marks	Average Performance	Marks	Good Performance	Marks	Total	PO Mapping	CO Mapping
Presentation (5 Marks)	Clarity and Organization of Presentation	Presentation lacks structure and clarity	0 – 0.5	Presentation moderately organized with minor clarity issues	1 – 1.5	Presentation is well-structured, clear, and professional	2	2	PO1, PO2, PO3, PO5, PO9, PO10	CO2
	Communication and Technical Explanation	Unable to explain project effectively	0 – 0.5	Basic explanation provided with moderate confidence	1	Technical concepts explained clearly and confidently	1.5	1.5	PO3, PO5, PO10, PO12	CO2
	Use of Visual Aids and Time Management	Poor use of slides/demo and improper time management	0	Adequate use of visuals with moderate time management	0.5	Effective use of visuals and excellent time management	1.5	1.5	PO5, PO8, PO10	CO2
Total								**5**		
Viva (10 Marks)	Understanding of Complete Project	Unable to explain project workflow and concepts	0 – 1	Basic understanding with partial explanations	1.5 – 2	Strong understanding of complete project workflow and concepts	3	3	PO1, PO2, PO3, PO4, PO5, PO10	CO2
	Knowledge of Design and Implementation	Unable to justify implementation choices	0 – 0.5	Partial justification provided	1 – 1.5	Implementation choices clearly justified with technical reasoning	2	2	PO2, PO3, PO4, PO5, PO8	CO2
	Response to Technical Questions	Unable to answer technical questions	0 – 0.5	Answers basic questions with support	1 – 1.5	Answers confidently with technical depth	2	2	PO1, PO2, PO10, PO12	CO2
	Individual Contribution and Team Coordination	Unable to explain individual contribution	0 – 0.5	Moderate contribution explained	1	Significant contribution and strong team coordination demonstrated	1.5	1.5	PO9, PO10, PO11	CO2
	Future Scope and Ethical Awareness	Unable to discuss future improvements or ethics	0	Basic future scope identified	0.5	Future enhancements and ethical considerations	1.5	1.5	PO6, PO7, PO8, PO12	CO2

						clearly explained				
Total								**10**		
Demo (5 Marks)	Functionality and Execution	Project fails to execute properly	0 – 0.5	Project executes partially with some issues	1	Project executes successfully with expected functionality	2	2	PO1, PO2, PO3, PO5	CO3
	Feature Integration and Output	Features incomplete and outputs incorrect	0 – 0.5	Basic features integrated with acceptable output	1	All major features integrated with accurate outputs	2	2	PO2, PO3, PO5, PO9	CO3
	Debugging and Stability	Frequent errors and instability	0	Minor issues observed during execution	0.5	Stable execution with proper debugging	1	1	PO4, PO5, PO12	CO3
Total								**5**		
Report (10 Marks)	Organization and Structure of Report	Report poorly structured and incomplete	0 – 0.5	Report adequately structured with minor gaps	1 – 1.5	Report professionally organized and complete	2	2	PO10, PO11, PO12	CO2, CO3
	Technical Content and Analysis	Insufficient technical explanation and analysis	0 – 0.5	Moderate technical content with basic analysis	1 – 1.5	Comprehensive technical explanation and detailed analysis	2	2	PO1, PO2, PO4, PO5, PO10	CO3
	Results and Discussion	Results not properly presented or analyzed	0 – 0.5	Basic result discussion provided	1 – 1.5	Results clearly presented with meaningful discussion	2	2	PO2, PO4, PO5, PO11	CO3
	References and Ethical Practices	Improper citation and plagiarism issues	0	References included with minor formatting issues	1	Proper citation format followed with originality maintained	1.5	1.5	PO8, PO10	CO2
	Documentation Quality with Similarity and AI Checking	High similarity index / excessive AI-generated content / poor formatting	0 – 0.5	Acceptable similarity percentage with moderate formatting quality	0.5 – 1	Low plagiarism, minimal AI dependency, and professional documentation quality	2.5	2.5	PO8, PO10, PO12	CO2, CO3
Total								**10**		

8. Direct Assessment Tools

Sl. No	Direct Assessment Tools		Weightage	CO1	CO2	CO3
1	Proposal Submission	Proposal	5		✓	
		LS	5	✓		
2	Design & Solution	Block diagram/architecture	5		✓	
		Clarity in writeup	5	✓		
3	Progress Review	Design to Code Mapping	6		✓	
		Code Explanation	14			✓
4	Mid Review	Presentation	10		✓	
		Viva	10			✓
		Result Analysis	10			✓
5	Final Review	Presentation	5		✓	
		Viva	10		✓	
		Demo	5			✓
6	Final Report	Content and Result presentation	10		✓	✓
Total			**100**			

9. Threshold and Target

CO	Threshold%	Target%
CO1	65	70
CO2	65	70
CO3	65	70
CO4	65	70

10. Faculty Signature

Faculty Name	Signature
Dr. Divya Udayan J [Mentor]	

Counter signed by

Class committee chair

Vice chair/Chair